

Lethal Company - Vollständige Übergabe bis S1.36

Maschinenlesbarer Projektstand in menschenlesbarer Form

Stand: 2026-09-02 | Lethal Company V81 | Canonical candidate: S1.36

**Wichtig: S1.36 ist Build-/Archiv-/Diff-verifiziert, aber noch nicht runtime-getestet.
Letzter Runtime-Test: S1.34.**

Dokumentzweck

Diese Fassung bündelt den aktuellen S1.36-Projektstand, Kernregeln, technische Baseline, offene Tests, Anti-Regression-Regeln, jüngste Arbeitsphase und Chronologie. Die Markdown-/TXT-/JSON-Dateien im Repository bleiben die primären Quellen für ChatGPT.

00 — Current State

Canonical project state: S1.36

Date: 2026-09-02

Current gameplay/test profile: Profiles/LC V1 S1.36 Handover Clean Baseline.r2z

Latest runtime-tested profile: Profiles/LC V1 S1.34 Malfunctions Disabled.r2z

Game: Lethal Company V81

Repository: <https://github.com/Tendas240/Lethal-Company-AI-Modding-Project>

S1.36 is the new canonical handover candidate and has passed archive/diff/build verification, but it has **not yet received a runtime test**. S1.34 is the most recent profile actually tested in game.

Read order for ChatGPT

1. START_HERE_ChatGPT_Masterprompt.txt
2. Current/00_CURRENT_STATE.md
3. Current/01_HANDOVER_CORE.md
4. Current/02_TECHNICAL_BASELINE.md
5. Current/04_OPEN_ISSUES_AND_NEXT_TESTS.md
6. Current/05_FAILED_AND_OBSOLETE_APPROACHES.md
7. Current/06_RECENT_WORK_S1.32-S1.36.md
8. Current/03_PROJECT_CHRONOLOGY.md
9. Current/Projektstatus_S1.36.json
10. Current/Aktive_Modliste_S1.36.txt
11. Current/S1.36_BUILD_VERIFICATION.txt
12. Current/VERIFIKATION_S1.36.txt

Then inspect Profiles/, Patches/, Logs/, and References/ as needed.

Historical S1.31 PDF/DOCX and versioned metadata belong under Archive/S1.31/ and must not be mistaken for current documentation.

Critical Gale import rule for S1.36

S1.36 contains a project-local DLL:

BepInEx/plugins/Tendas-S135CompatibilityFixes/S135CompatibilityFixes.dll

When importing the .r2z in Gale, enable:

Advanced options → Import all files

Otherwise Gale may skip the embedded DLL. If that happens, import this local-mod package separately:

Patches/S135CompatibilityFixes/Tendas-S135CompatibilityFixes-1.0.0.zip

A valid S1.36 runtime must log both:

- S1.35 Compatibility Fixes loaded
- [EnemyScanFix] Patched EnemyScan to list every active EnemyAI ...

If these lines are absent, do not evaluate the ship-door or complete-enemy-list fixes; the local patch was not loaded.

Priority rule

The chronologically newest confirmed runtime/profile fact overrides older assumptions. Runtime evidence overrides package-manifest assumptions when they disagree.

Archive/ is historical reference material and must not override a newer confirmed state unless current documentation explicitly refers back to it.

Current build lineage

S1.29 gameplay base → S1.30 → S1.31 → S1.32 → S1.33 → S1.34 → S1.35 → S1.36

S1.29D was a diagnostic derivative of S1.29 and is never a gameplay base.

Recent distinctions:

- **S1.32:** Leaf Boy added to the existing LethalMin Attack Blacklist; Mirage recording retention enabled.
- **S1.33:** first custom ship-door failsafe attempt; AJB door mod disabled.
- **S1.34:** Malfunctions disabled by explicit user decision. Runtime proved the custom S1.33 DLL had not been imported/loaded, so door behavior fell back to vanilla.
- **S1.35:** rebuilt local patch for ship-door failsafe + complete EnemyScan output; structurally verified but not runtime-tested.
- **S1.36:** S1.35 plus one cleanup: ProjectSCP-SCP999 disabled because current logs proved it was still loading and throwing a startup NRE despite older documentation saying it was disabled.

Explicitly disabled in S1.36

Manifest: 176 entries, 170 active, 6 disabled.

- AJB-Keep_hangar_ship_door_closed
- zealsprince-Malfunctions

- Reiko88-Observer
- ProjectSCP-SCP999
- Kittenji-Dont_Touch_Me
- SoftDiamond-BrutalCompanyMinusExtraReborn

Malfunctions must remain disabled until the user explicitly asks for it to be re-enabled.

Confirmed working / stable from earlier runtime tests

- Hold-to-Scan via LethalHUD.
- Pikmin water resistance.
- Deadline 0 → Company/Gordion routing and automatic landing via CompanyBuildingEnhancements.
- 26 normal interiors at equal Weight 100 including Black Mesa.
- Rolling Giant / Shy Guy / Siren Head native spawn ownership.
- RandomEnemiesSize.
- GeneralImprovements quota rollover.
- x753-Mimics and CoronerMimics removed.
- CodeRebirth Flash Turret no longer affects Pikmins.
- Mirage neverDeleteRecordings=true is carried forward.

Leaf Boy attack-loop diagnosis is confirmed; S1.32+ blacklists Leaf boy. Runtime confirmation that the blacklist fully stops the loop is still desirable.

Important recent findings

- The S1.32 ship-door close trigger itself was not conclusively identified. Malfunctions had no successful relevant door event in the analyzed run.
- A Masked enemy was nearby, but vanilla Masked AI has no hangar-button interaction. Poltergeist dead-player/ghost interaction is a separate mechanic.
- The permanent-lockout risk came from an external/normal close combined with unconditional door-power refill.
- S1.34 proved our custom door DLL had not loaded; its countdown/opening was vanilla behavior, not a failed test of the new algorithm.
- EnemyScan 1.2.1 intentionally omitted EnemyAI without ScanNodeProperties; the S1.35/S1.36 patch removes that display-only filter.
- Puma / PumaAI is the vanilla V80+ Feiopar, not a mod enemy.
- Coin is CodeRebirth currency collected/stored through the Denomination Analyzer and used by CodeRebirth merchant/vending systems.
- Current runtime logs showed ProjectSCP-SCP999 2.4.0 loading and throwing NullReferenceException in SCP999.Plugin.Awake(); S1.36 corrects the manifest state to disabled.

Immediate next test

Import S1.36 with Import all files and verify the local patch loads before testing anything else.

Then:

1. Close the hangar door while the only living player is inside the landed ship: power should remain at 100%.

2. Close the door with all living players outside: vanilla hydraulic countdown should run and reopen the door.
3. Use enemies while known scanless/modded enemies are active and compare against visible/runtime-spawned enemies.
4. Confirm no SCP999 plugin load or startup NRE occurs.
5. Confirm Leaf Boy is no longer attacked by Pikmins.

After the run, preserve the full LogOutput.log.

01 — Handover Core

Working rules for the next ChatGPT conversation

1. Treat **S1.36** as the current canonical handover/test candidate unless the repository contains a newer confirmed state.
2. Treat **S1.34** as the latest runtime-tested profile until S1.36 is actually run.
3. Never treat S1.29D as a normal gameplay base; it existed only for the RedPill enemy-power audit.
4. Before changing enemy spawning, determine the actual spawn owner for the enemy.
5. Do not independently distort the binding screenshot reference ratios.
6. Do not modify the confirmed 26×Weight100 interior architecture without a concrete runtime reason.
7. Unknown enemy PowerLevels must not be guessed.
8. Do not reintroduce solutions listed in 05_FAILED_AND_OBSOLETE_APPROACHES.md without new evidence.
9. **Malfunctions remains disabled until the user explicitly requests re-enabling it.**
10. **ProjectSCP-SCP999 remains disabled.** Current logs proved it was accidentally active and throwing a startup NRE.
11. Do not enable AJB-Keep_hangar_ship_door_closed alongside the S1.35/S1.36 local patch.
12. For S1.36, verify the local plugin actually loads before judging door or EnemyScan behavior.

Critical S1.36 import/install rule

The current profile embeds S135CompatibilityFixes.dll. Gale can skip extra arbitrary files on a normal profile import.

Use **Advanced options → Import all files.**

If the plugin does not appear in the BepInEx startup log, import this local mod separately into the same profile:

Patches/S135CompatibilityFixes/Tendas-S135CompatibilityFixes-1.0.0.zip

Expected startup evidence:

- S1.35 Compatibility Fixes loaded
- [EnemyScanFix] Patched EnemyScan to list every active EnemyAI regardless of ScanNodeProperties.

Absence of those messages means the fix is not installed.

Gameplay-base lineage

S1.29

LC V1 S1.29 CodeRebirth Runtime Test.r2z

Normal gameplay base with CodeRebirth 1.6.9.

S1.29D

LC V1 S1.29D Enemy Power Audit.r2z

Diagnostic only. Never use as a gameplay base.

S1.30

LC V1 S1.30 Power Caps Mimicless Pikmin Shield.r2z

Built from S1.29, not S1.29D.

Major changes:

- x753-Mimics removed.
- CoronerMimics removed.
- CodeRebirth↔Pikmin hazard switches set false.
- Indoor caps set to the then-selected higher values.
- Enemy weights and 26-interior rotation preserved.

S1.31

LC V1 S1.31 Indoor Power Trim -4.r2z

All controllable indoor power caps reduced by 4. No intended weight/interior/Pikmin-protection changes.

S1.32

LC V1 S1.32 Leaf Boy Blacklist + Mirage Keep Recordings.r2z

Changes:

- appended Leaf boy to the existing LethalMin Attack Blacklist;
- added Mirage game-root settings with neverDeleteRecordings=true while preserving the other observed Mirage values.

Runtime finding: the player returned from the dungeon to a closed ship hangar door and could be locked out indefinitely because the old AJB mod prevented the hydraulic countdown from recovering the door.

S1.33

LC V1 S1.33 Ship Door Failsafe.r2z

First custom replacement for AJB:

- AJB disabled;
- custom DLL intended to freeze power only while a living player was inside;
- vanilla drain intended to resume when all living players were outside;
- DoorAudit logging added.

Later runtime evidence showed this DLL was not imported/loaded by Gale, so the algorithm was not actually tested.

S1.34

LC V1 S1.34 Malfunctions Disabled

Latest runtime-tested profile.

Change: zealsprince-Malfunctions disabled by explicit user decision and intended to stay disabled until the user explicitly asks otherwise.

Runtime findings:

- custom S1.33 door plugin did not load;
- no DoorAudit/DoorFailsafe output;
- door energy drained and reopened at zero = vanilla behavior;
- EnemyScan omitted some spawned enemies;
- Puma was active and is vanilla Feiopar;
- Coin came from CodeRebirth;
- SCP999 still loaded and threw a startup NRE.

S1.35

LC V1 S1.35 Door + Complete Enemy Scan.r2z

Structurally verified but not runtime-tested.

Rebuilt local patch:

- ship-inside detection using isInHangarShipRoom plus shipInnerRoomBounds fallback;
- door audit logging;
- anti-lockout behavior that allows vanilla drain only when all living players are outside;
- EnemyScan output patch that lists all active EnemyAI rather than requiring ScanNodeProperties.

S1.36

LC V1 S1.36 Handover Clean Baseline.r2z

Current canonical candidate.

Exact intended delta from S1.35:

- profile name updated;
- ProjectSCP-SCP999 set enabled:false;
- no other archive member changed.

S1.36 carries all S1.35 fixes plus the S1.34 Malfunctions-disabled policy.

Stable architecture

Spawn ownership

Use one positive spawn owner per enemy wherever practical.

Especially:

- Rolling Giant → native mod configuration.
- Shy Guy / Scopophobia → native mod configuration.
- Siren Head → native mod configuration.

Do not force them positive through LethalLevelLoader again without new evidence.

Binding enemy-weight references

The seven screenshots under References/Spawn-Rates/ are binding for relative ratios among the enemies shown on Experimentation, Assurance, Vow, Offense, March, Adamance and Titan.

New enemies may enlarge the total pool, but reference enemies must not be independently rebalanced against each other without a deliberate new decision.

Interiors

26 normal dungeon flows rotate equally at Weight 100. Black Mesa uses its own DawnLib/config path and must not be double-registered through LLL.

Company automation

Confirmed solution: CompanyBuildingEnhancements 2.6.0.

Do not return to AutoCompanyBuilding or RandomMoonFX for this purpose.

Pikmin protection

Core settings:

- No Knock Back = true
- Invinceable Pikmin = true
- Pikmin Die In Player Death Zones = false

S1.30 CodeRebirth compatibility switches false:

- ACU Targets Winged Pikmin
- ACU Bullet Knockbacks Pikmin
- Crane Targets Pikmin
- Crane Squishes Pikmin
- Fan Knockbacks Pikmin

- Microwave Knockbacks Pikmin
- Flash Turret Knockbacks Pikmin
- Laser Turret Kills Pikmin
- Tornado Pulls Pikmin
- Compactor Squishes Pikmin

Confirmed: Flash Turret protection works.

S1.31 Leaf Boy issue: Pikmins attacked LeafBoi(Clone) continuously for several minutes.
S1.32+ appends Leaf boy to the current Attack Blacklist.

Still unconfirmed:

- Microwave burn/cook immunity;
- complete enemy-AI target exclusion for immortal Pikmins;
- remaining CodeRebirth hazards;
- runtime confirmation that the Leaf Boy blacklist fully stops the loop.

Ship-door facts

- The original S1.32 close trigger was not conclusively identified.
- Malfunctions did not show a successful relevant door malfunction in the analyzed run.
- BCMER was disabled.
- A Masked enemy was nearby, but vanilla MaskedPlayerEnemy has no StartButton/StopButton/HangarShipDoor interaction.
- Poltergeist can allow dead players/ghosts to interact with ship-door buttons; this is distinct from Masked AI.
- Permanent lockout came from a close action combined with unconditional AJB door-power refill.
- S1.35/S1.36 replaces AJB with a narrower failsafe and reuses vanilla hydraulic drain/open logic when everyone living is outside.

EnemyScan facts

EnemyScan 1.2.1 originally used a ScanNodeProperties filter in BuildEnemyCountString(), so spawned enemies without a scan node could be omitted from enemies.

S1.35/S1.36 patches only that list-building output. It does not change spawn rates, PowerLevels, AI, scan nodes, or bestiary data.

Mirage facts

Current settings:

- localPlayerVolume = 0.5
- neverDeleteRecordings = true
- allowRecordVoice = true
- muteVoiceMimic = false

Mirage stores:

- settings: <Lethal Company installation folder>/Mirage/settings.json
- recordings: <Lethal Company installation folder>/Mirage/Recording

This is game-root storage, not BepInEx/config.

Important enemy/system decisions

- SCP-999 stays disabled.
- Malfunctions stays disabled until explicit user instruction.
- Immortal Snail maximum: 2 simultaneously.
- Observer remains disabled.
- Don't Touch Me remains disabled.
- BCMER remains disabled but is not permanently obsolete.
- CodeRebirthLib must not be installed.
- Peepers enemy/hazard remains removed; do not confuse it with the Peeper tool.
- LethalPlaytime Boxy Boo / Huggy Wuggy / Miss Delight should not be reactivated on V81.

Current unresolved enemy PowerLevels

Do not invent values for:

- Rolling Giant
- Siren Head
- Immortal Snail
- Herobrine
- Football
- Faceless Stalker
- CodeRebirth Debt Collector / Boogey Man

02 — Technical Baseline

Current manifest: S1.36

Profile:

Profiles/LC V1 S1.36 Handover Clean Baseline.r2z

- 176 Thunderstore manifest entries
- 170 active
- 6 explicitly disabled
- plus one required project-local compatibility plugin embedded in the archive and available separately under Patches/

Explicitly disabled:

- AJB-Keep_hangar_ship_door_closed 1.0.0
- zealsprince-Malfunctions 1.10.3
- Reiko88-Observer 2.0.1
- ProjectSCP-SCP999 2.4.0
- Kittenji-Dont_Touch_Me 1.2.8
- SoftDiamond-BrutalCompanyMinusExtraReborn 1.71.0

The full package list is in Current/Aktive_Modliste_S1.36.txt.

Required local plugin

Patches/S135CompatibilityFixes/

Package:

Patches/S135CompatibilityFixes/Tendas-S135CompatibilityFixes-1.0.0.zip

Embedded DLL inside S1.36:

BepInEx/plugins/Tendas-S135CompatibilityFixes/S135CompatibilityFixes.dll

Functions:

1. ship-door anti-lockout behavior,
2. door audit logging,
3. EnemyScan complete-list patch.

Gale import requirement

Use **Advanced options → Import all files** when importing S1.36.

If BepInEx does not log S1.35 Compatibility Fixes loaded, import the local-mod ZIP manually into the same profile.

Package versions vs runtime plugin versions

The export lists Thunderstore package versions. Current logs prove that some packages contain assemblies whose internal plugin versions differ.

For code/error analysis, prefer the newest runtime log over the manifest when they disagree.

Examples from current runs:

- SirenHead package 2.0.7 → runtime plugin 2.0.3
- ImmortalSnailFork package 0.1.1 → runtime plugin 0.1.0
- Black Mesa package 3.4.4 → runtime plugin 3.4.1
- Coroner package 2.4.1 → runtime plugin 2.3.0
- darmuhsTerminalStuff package 3.10.2 → runtime plugin 3.10.1
- Lethal Doors Fixed package 1.2.1 → runtime plugin 1.2.0
- FairAI package 1.6.1 → runtime plugin 1.6.0

Do not silently treat these as package corruption; package version and assembly/plugin version can legitimately differ. Always state which source is being used.

Indoor power caps

Unchanged from S1.31.

Vanilla

Moon	Indoor Power
Experimentation	4
Assurance	8
Vow	10

Moon	Indoor Power
Embrion	12
Rend	16
Dine	16
Offense	20
Adamance	22
Artifice	22
Liquidation	22
March	24
Titan	32
Gordion / Company	0

These came from the explicit S1.30-values-minus-4 decision. Do not rebalance them while testing unrelated fixes.

Custom/external

Moon	Indoor Power
EGypt	20
PsychSanctum	18
Abaddon	36
Cabal	20
Arcadia	16
Argent	18
Zenit	18
Lament	10
Flicker	14
Shutter	0
Pareidolia	36
Vigilance	18
Bozoros	32
Sanguine	20
Spectralis	22
The Iris	36
Black Mesa	28
Oxyde	not separately controllable in current export

Do not guess an Oxyde value.

26 equal-weight interiors

All intended at Weight 100 on normal moons:

1. Facility
2. Haunted Mansion
3. Mineshaft
4. Bunker
5. Drains
6. Substation

7. Liminal Facility
8. Storehouse
9. Tower
10. Circus Facility
11. Atlantean Citadel
12. Deep Sewers
13. Fractured Complex
14. Greenhouse
15. Art Gallery
16. Decrepit store
17. Expanded facility
18. Expanded Mineshaft
19. Grand Armory
20. Spooky manor
21. ScoopyCastle
22. Slaughterhouse
23. Storage Complex
24. Rubber Rooms
25. Toy Store
26. Black Mesa

Runtime previously confirmed all 26 viable at Weight 100 with an empty unviable list.

Black Mesa uses its own DawnLib/config path; do not double-register through LLL.

LethalMin baseline

Core Pikmin settings:

- No Knock Back = true
- Invincible Pikmin = true
- Pikmin Die In Player Death Zones = false

Current S1.36 Attack Blacklist line:

Docile Locust Bees,Manticoil,Red Locust
Bees,Blob,Nemo,InternNPC,BellCrab,Nancy,Transporter,Janior,Peace
Keeper,Guardman,Tornado,FireStorm,Hurricane,Cabinet, Leaf boy

Do not replace this with an older shorter list. S1.32 appended Leaf boy to the existing current list.

Mirage baseline

S1.36 carries:

localPlayerVolume=0.5

neverDeleteRecordings=true

allowRecordVoice=true

muteVoiceMimic=false

Mirage paths:

- settings: <Lethal Company installation folder>/Mirage/settings.json
- recordings: <Lethal Company installation folder>/Mirage/Recording

neverDeleteRecordings=true is intentional and must remain unless the user asks otherwise.

Ship-door compatibility patch

The S1.35/S1.36 local plugin patches HangarShipDoor.Update() with a late postfix.

Behavior only while landed and hangar closed:

- living player inside ship → doorPower forced back to 1f after vanilla Update, so power stays 100%;
- living players exist but all are outside → no refill, vanilla hydraulic drain continues and opens the door at zero;
- no living controlled players, ship leaving/orbit/ship phase, or open door → no intervention.

Inside detection uses both:

- player.isInHangarShipRoom
- StartOfRound.shipInnerRoomBounds.bounds.Contains(player.transform.position) fallback

Diagnostics:

- [DoorAudit]
- [DoorFailsafe]

The patch also logs stack traces for SetDoorClosed, SetDoorOpen, PlayDoorAnimation and Start/Stop ship-door button interactions.

AJB-Keep_hangar_ship_door_closed is disabled to avoid competing postfixes.

EnemyScan complete-list patch

Original EnemyScan 1.2.1 only listed EnemyAI that had a ScanNodeProperties child.

S1.35/S1.36 patches only BuildEnemyCountString() so that it groups every active EnemyAI with a valid EnemyType by enemyType.enemyName.

It does not modify:

- spawn weights,
- spawn caps,
- PowerLevels,
- AI,
- ScanNodes,
- bestiary registration.

Coin / CodeRebirth currency

Coin is CodeRebirth content, namespaced as code_rebirth:coin.

CodeRebirth Money.TryCollectCoin():

- requires a MoneyCounter supplied by the Denomination Analyzer;
- without it, displays a hint telling the player to buy the analyzer from the ship terminal;
- with it, adds the coin value to stored CodeRebirth currency and destroys/despawns the coin.

The stored currency is used by CodeRebirth merchant/vending systems.

Puma / Feiopar

PumaAI is the internal vanilla AI class for the V80+ enemy Feiopar. Puma log lines do not imply a Puma mod is installed.

SpawnCycleFixes

Spawn Cycle Fixes 1.2.2 remains active.

Approximate consistent spawn waves historically identified:

- 07:39
- 09:00
- 11:00
- 13:00
- 15:00
- 17:00
- 19:00
- 21:00
- 23:00

Do not disable it blindly. If enemies feel too late, investigate probability/amount curves rather than simply removing SpawnCycleFixes.

Known current warnings/issues

SCP999 startup NRE

S1.31–S1.34 logs showed Loading [SCP999 2.4.0], immediately followed by NullReferenceException in SCP999.Plugin.Awake().

S1.36 disables the package. Next runtime must verify the load/NRE is gone.

SellMyScrap

Current logs show warnings because several methods cannot load ShipInventoryUpdated, Version=2.0.0.0. SellMyScrap otherwise continues loading. Investigate only if related functionality is broken.

InjectionLibrary native DLL warnings

Mirage/Opus native DLLs are skipped as non-.NET assemblies by InjectionLibrary. Those scanner warnings alone are not a failure.

CodeRebirth Weather Registry

Weather Registry not found, skipping weather content registration appears while CodeRebirth otherwise continues loading. Treat as an open compatibility warning, not a proven blocker.

NavMeshInCompany

Missing NodeHelper script warnings remain known. Reassess only if Company navigation is actually broken.

04 — Open Issues and Next Tests

Priority 0 — Verify S1.36 import

S1.36 is structurally verified but not runtime-tested.

When importing in Gale, enable:

Advanced options → Import all files

Then check LogOutput.log for:

- S1.35 Compatibility Fixes loaded
- [EnemyScanFix] Patched EnemyScan to list every active EnemyAI ...

If missing, import:

Patches/S135CompatibilityFixes/Tendas-S135CompatibilityFixes-1.0.0.zip

into the S1.36 profile.

Do not evaluate door/EnemyScan behavior until this load requirement is satisfied.

Priority 1 — Ship-door behavior

Test A — living player inside ship

As the only living player, stand clearly inside the landed ship and close the hangar door.

Expected:

- door closes;
- energy remains at 100%;

- door stays closed indefinitely until intentionally opened or another legitimate game transition occurs.

Log should show DoorAudit and livingInsideShip=1.

Test B — all living players outside

With all living players outside, reproduce a closed hangar door.

Expected:

- local patch does not refill energy;
- vanilla hydraulic countdown runs;
- door opens at zero;
- permanent outside lockout cannot occur.

Log should show DoorFailsafe saying all living players are outside.

If the door closes unexpectedly again, preserve the new audit stack because it may identify the calling method/mod.

Priority 2 — Complete enemies output

EnemyScan 1.2.1 originally excluded EnemyAI without ScanNodeProperties.

S1.36 patches only this listing behavior.

Test while several known enemies are active, ideally including a modded or scanless case.

Compare:

- visible enemies,
- runtime spawn/AI lines,
- enemies terminal output.

Goal: every active EnemyAI appears and counts update correctly.

Priority 3 — Verify SCP999 is truly gone

Latest S1.31-S1.34 logs showed SCP999 2.4.0 loading and immediately throwing a startup NRE.

S1.36 disables ProjectSCP-SCP999.

Expected:

- no Loading [SCP999 ...];
- no SCP999.Plugin.Awake NRE;
- no later [SCP999] Max health ... debug lines.

If it still loads, investigate whether Gale ignored the disabled state or another copy exists outside the profile.

Priority 4 — Leaf Boy blacklist verification

S1.32+ appends Leaf boy to the LethalMin Attack Blacklist.

Original issue: multi-minute Pikmin attack loop against LeafBoi(Clone).

Confirm:

- Pikmins do not start attacks on Leaf Boy;
- no repeated LethalMin hit loop occurs;
- Leaf Boy otherwise behaves normally.

Do not alter Leaf Boy spawn rate/health to solve targeting.

Priority 5 — CodeRebirth Microwave vs Pikmins

Microwave Knockbacks Pikmin=false does not prove burn/cook immunity.

Still unproven:

- burn immunity;
- cook/status immunity;
- prevention of permanent burning while Pikmins are immortal.

If persistent burning occurs, config is insufficient and a targeted compatibility patch is required.

Priority 6 — Enemy targeting of immortal Pikmins

Immortality does not prove enemies stop selecting Pikmins as targets.

Watch for enemies remaining occupied with unkillable Pikmins.

If reproducible, prefer a small target-filter compatibility patch over disabling enemies.

Priority 7 — Other CodeRebirth hazards

Flash Turret is confirmed safe.

Still test when encountered:

- ACU
- Crane
- Fan
- Laser Turret
- Tornado
- Compactor

Priority 8 — SellMyScrap warnings

Current logs contain warnings that some SellMyScrap methods cannot load ShipInventoryUpdated 2.0.0.

Do not change anything solely because of the warning. First determine whether user-facing SellMyScrap functionality is actually broken.

Priority 9 — Beehives

Do not use Offense as the bee baseline.

Prefer Assurance or March.

Only compensate Bee chance if several runs on actual vanilla Bee moons show a reproducible problem.

Priority 10 — Dungeon theme song

Main candidates:

1. Haunted Harpist / Phantom Piper — spatial enemy music.
2. PizzaTowerEscapeMusic — event/escape music, likely tied to Apparatus behavior.

Record moon, interior, time, Apparatus state, directionality and enemies output when it occurs.

Priority 11 — Exact enemy PowerLevels

Still unresolved:

- Rolling Giant
- Siren Head
- Immortal Snail
- Herobrine
- Football
- Faceless Stalker
- CodeRebirth Debt Collector / Boogey Man

Do not guess. Use runtime or binary/asset analysis.

Lower-priority observations

- One disabled vanilla Turret was observed historically. Investigate only if repeated with player/terminal/hacking actions ruled out.
- CodeRebirth logs Weather Registry not found; investigate only if missing weather content matters.
- NavMeshInCompany missing NodeHelper warnings are known; investigate if navigation is actually broken.

After every meaningful run

Preserve the full LogOutput.log and compare at least:

- exceptions/NREs;
- plugin load list;
- DoorAudit / DoorFailsafe;
- EnemyScanFix;
- SCP999 absence;
- Pikmin hazard interactions;
- LethalMin work-state warning volume;
- interior viability/weights;
- load/landing anomalies.

05 — Failed, Obsolete, or Deliberately Parked Approaches

Do **not** reintroduce the following without new technical evidence or explicit user instruction.

Component / approach	Status / reason
LethalCompanyVariables / EnemyRarityConfig	Historically removed; do not restore as the spawn solution.
LethalQuantities 1.2.9 as global enemy owner	Enemy-control sections examined were not the active source of truth.
AutoCompanyBuilding 1.2.1	Loaded but did not route reliably under V81.
AutoCompanyBuilding 1.1.3	Routing could work, automatic landing did not.
RandomMoonFX 1.4.4 for Company automation	Active state interfered with manual moon selection; disabled state lacked needed routing patch.
Old Hold_Scan_Button	Replaced by working LethalHUD Hold-to-Scan.
Peepers enemy/hazard mod	Removed. Do not confuse with the Peeper tool.
CodeRebirthLib	Deprecated/unwanted. Do not install.
ProjectSCP-SCP999	Must remain disabled. Recent S1.31–S1.34 logs proved it was accidentally active and throws a startup NRE in SCP999.Plugin.Awake(). S1.36 fixes the manifest.
Gnomes	Caused V81 PlayerIsTargetable MissingMethod spam; removed.
FacilityMeltdown	Fully removed.
ASTeam Racist Hoarding Bugs sound replacer	Removed.
FearOverhauled	Removed.
LethalModDataLib	Removed after ShipWindows update; old NRE disappeared.
LethalPlaytime Boxy Boo / Huggy Wuggy / Miss Delight	Do not reactivate on V81 due to AI/collision problems.
MirageClipLimiter.dll	Unrealized plan; DLL does not exist.
Forcing Rolling Giant / Shy Guy / Siren Head positive weights through LLL	Native ownership is the reliable architecture.
AJB-Keep_hangar_ship_door_closed as current door solution	Disabled in S1.33+. Its unconditional power refill can turn an external close into a permanent outside lockout. Replaced by the narrower local failsafe.
Treating the S1.34 door countdown as proof the S1.33 algorithm failed	Invalid conclusion. The custom DLL never loaded; S1.34 was vanilla door behavior after AJB was disabled.
EnemyScan 1.2.1 original scan-node-filtered output as a complete enemy census	Incomplete by design. Base mod remains active, but S1.35/S1.36 patches its list-building method.
S1.35 as canonical profile	Superseded during handover because it still had ProjectSCP-SCP999 enabled. S1.36 is S1.35 plus that correction.

Deliberately disabled / parked, not permanently obsolete

Malfunctions

zealsprince-Malfunctions is disabled from S1.34 onward by explicit user decision.

Do not re-enable it unless the user explicitly says to do so in the future.

This is a persistent project policy, not a claim that every Malfunctions feature is broken.

BCMER

SoftDiamond-BrutalCompanyMinusExtraReborn remains disabled in the current baseline. It is not permanently forbidden.

If the user later wants it back, re-audit the current version/config and integrate it from the correct gameplay base without letting it silently override established spawn power/chance ownership.

General anti-regression rule

Before proposing an old mod or configuration as a fix, search this file, 06_RECENT_WORK_S1.32-S1.36.md and the chronology first.

A solution that was previously rejected must not be presented as new unless there is a concrete reason why the old failure mode no longer applies.

06 — Recent Work: S1.32 to S1.36

This file preserves the detailed handover facts from the work phase after S1.31 so diagnostic context is not lost.

S1.31 Leaf Boy / LethalMin incident

User observation:

- Pikmins attacked Leaf Boys continuously.
- Leaf Boy appeared unable to end the interaction.
- Because Pikmins are immortal, the engagement could continue indefinitely.

Runtime evidence showed:

- LethalMin registered Leaf boy as a Pikmin enemy.
- LethalConfig registered the LethalMin/Pikmin Behavior/Attack Blacklist setting.
- repeated Pikmin hits against LeafBoi(Clone),
- repeated real damage aggregation through LethalMin over several minutes.

Historical handover data explicitly warned not to replace the newer longer blacklist with an old shorter list.

Decision:

- append exactly Leaf boy to the existing Attack Blacklist;
- do not change Leaf Boy spawn chance, health, or other behavior merely to solve Pikmin targeting.

S1.36 still carries this append-only blacklist change.

Mirage recording retention

Observed runtime settings before the change:

- localPlayerVolume: 0.5
- neverDeleteRecordings: false
- allowRecordVoice: true
- muteVoiceMimic: false

Mirage source/documentation established game-root storage:

- settings: <Lethal Company folder>/Mirage/settings.json
- recordings: <Lethal Company folder>/Mirage/Recording

Decision in S1.32+:

- neverDeleteRecordings=true;
- preserve the other observed settings.

S1.32 unexpected ship-door lockout

User test:

- left the ship for the dungeon;
- returned to a closed hangar door;
- could not enter;
- AJB Keep hangar ship door closed prevented hydraulic power from counting down, so the door never auto-reopened.

Investigation:

- Malfunctions was active at that time, but the analyzed run did not show a successful relevant door malfunction.
- BCMER was disabled.
- a MaskedPlayerEnemy was near the ship around a suspicious time window.
- source inspection of vanilla MaskedPlayerEnemy found no interaction with StartButton, StopButton, or HangarShipDoor controls.
- Poltergeist can let dead players/ghosts interact with ship-door buttons; that is distinct from Masked AI.
- vanilla logging does not reliably identify the actor/source for an ordinary hangar-button close.

Conclusion:

- exact original close trigger remained unproven;
- permanent lockout was clearly caused by a close action combined with AJB's unconditional door-power refill.

S1.33 first door-failsafe design

Goal:

- retain unlimited closed-door power while a living player is inside;

- prevent permanent outside lockout by allowing vanilla hydraulic drain when every living player is outside;
- log door state/method stacks for future attribution.

Implementation:

- AJB disabled;
- custom S133ShipDoorFailsafe.dll added.

Important later discovery:

- the DLL was embedded in the .r2z but normal Gale import did not install/load it;
- therefore S1.33's algorithm was never actually runtime-tested.

Malfunctions policy from S1.34

The user explicitly requested that zealsprince-Malfunctions be disabled and remain disabled until the user might explicitly request reactivation in the future.

This is a persistent project decision.

S1.36 keeps it disabled.

S1.34 runtime test — door behavior

User test while alone inside the landed ship:

- closed hangar door from inside;
- door energy percentage decreased;
- at 0% the door reopened.

Log audit:

- no S1.33 Ship Door Failsafe plugin load line;
- no DoorAudit;
- no DoorFailsafe.

Conclusion:

- custom DLL was not installed/loaded by Gale;
- AJB was also disabled;
- observed behavior was pure vanilla hydraulic behavior, not failure of the custom algorithm.

Operational rule for future local-DLL profiles:

- Gale Advanced options → Import all files, or
- import the supplied local-mod ZIP separately.

S1.34 runtime test — EnemyScan

User suspected the enemies terminal command did not show every spawned enemy.

EnemyScan 1.2.1 source confirmed the reason in BuildEnemyCountString():

```
FindObjectsOfType<EnemyAI>().Where(ai =>  
ai.GetComponentInChildren<ScanNodeProperties>() is not null)
```

The mod intentionally hid enemies that were not scannable / lacked a scan node.

Conclusion:

- enemies was not a complete active-enemy census.

S1.35/S1.36 patches only that list-building result so all active EnemyAI with valid EnemyType are grouped and shown.

S1.34 runtime test — Coin

Log/runtime registration showed Coin under CodeRebirth.

CodeRebirth source confirms:

- Coin is code_rebirth:coin;
- collecting it requires a MoneyCounter supplied by the Denomination Analyzer;
- without the analyzer the player gets a hint to buy it from the ship terminal;
- when collected, the coin adds its value to CodeRebirth stored currency and despawns;
- stored money is used by CodeRebirth merchant/vending systems.

S1.34 runtime test — Puma

Puma / PumaAI is not a mod name.

It is the internal vanilla class/name for the V80+ Feiopar enemy.

Other mods may patch PumaAI, but the enemy itself is vanilla.

S1.35 rebuilt compatibility plugin

S135CompatibilityFixes.dll was compiled against V81 GameLibs.

Door logic:

- counts living controlled players only;
- inside status uses isInHangarShipRoom plus shipInnerRoomBounds fallback;
- refills door power only when landed, closed, and at least one living player is inside;
- when all living players are outside, vanilla drain/open logic is allowed;
- orbit/leaving/no-living-player states are left alone;
- door method/button audit logging retained.

EnemyScan patch:

- no ScanNode requirement for terminal listing;
- no spawn/AI/Power changes.

A standalone local-mod ZIP was created as installation fallback.

S1.35 passed compile/archive/diff checks but was not runtime-tested.

SCP999 contradiction found during handover

Older current documentation claimed SCP999 was disabled after the historical NRE incident.

Current S1.31, S1.32 and S1.34 runtime logs explicitly showed:

- Loading [SCP999 2.4.0]
- immediate NullReferenceException in SCP999.Plugin.Awake()
- later SCP999 Max health debug lines.

The S1.35 manifest still had ProjectSCP-SCP999 enabled:true.

Handover decision:

- do not freeze this known regression into the new baseline;
- create S1.36 as S1.35 plus exactly ProjectSCP-SCP999 enabled:false.

S1.36 canonical handover candidate

Profile:

Profiles/LC V1 S1.36 Handover Clean Baseline.r2z

Manifest:

- 176 total
- 170 active
- 6 disabled

Disabled:

- AJB-Keep_hangar_ship_door_closed
- zealsprince-Malfunctions
- Reiko88-Observer
- ProjectSCP-SCP999
- Kittenji-Dont_Touch_Me
- SoftDiamond-BrutalCompanyMinusExtraReborn

Archive verification confirmed:

- only export.r2x changed relative to S1.35;
- custom S135CompatibilityFixes DLL remains embedded;
- Leaf boy blacklist remains;
- Mirage neverDeleteRecordings=true remains;
- no other S1.35 archive member changed.

Runtime status:

not yet tested.

The next chat should begin with the controlled S1.36 import/load/door/enemy-list/SCP999 tests in 04_OPEN_ISSUES_AND_NEXT_TESTS.md.

03 — Project Chronology

This is a technical reconstruction of the confirmed project history. It is not claimed to be a verbatim transcript.

S1.2 — Profilename/Path-Length-Stabilisierung

Ein zu langer Gale-Profilname führte zu Windows-/BepInEx-Pfadproblemen. Profilename verkürzt; Coroner/Logging stabilisiert.

Result / significance: Dauerregel: Profilnamen kurz halten.

S1.3 — Frühe Gameplay-/Pikmin-/Enemy-Bereinigung

Wild-Pikmin-Falloff und unerwünschte Skalierung neutralisiert; Stamina angepasst; Football-Spawns über eigene Rarities auf 0; BCMER deaktiviert.

Result / significance: Mehrere Spawn-Systeme als konkurrierende Owner erkannt.

S1.4-S1.5 — Audio-/Kompatibilitätsdiagnose

Mirage-/Voice-Mimicking untersucht. Geplanter 20-Sekunden-MirageClipLimiter konnte nicht gebaut werden.

Result / significance: MirageClipLimiter.dll existiert nicht.

S1.6 — Pikmin- und Battery-Tuning

Indoor Pikmin 0.08→0.12; historische Battery-Anpassungen; BCMER weiter aus.

Result / significance: Battery-Lösung später ersetzt.

S1.7 — Facility-/Performance-Phase

Stamina testweise 2×; Facility-Verteilung; FacilityMeltdown entfernt; Rolling Giant temporär wegen Spam-Verdacht deaktiviert.

Result / significance: Rolling Giant war nicht die eigentliche Spam-Ursache.

S1.8 — Gnome-Ursache isoliert

Gnomes als V81 PlayerIsTargetable MissingMethod-Spamquelle entfernt. Rolling Giant zurück. Stamina ~1.5× finalisiert. MoreBattery eingeführt.

Result / significance: Gnomes nicht wieder hinzufügen.

S1.9 — Pikmin/Peepers Cleanup

Indoor Pikmin 0.12→0.11; Peepers vollständig entfernt.

Result / significance: Peeper-Tool ist davon getrennt.

S1.10 — Spawn-Autoritäten-Audit

Indoor Pikmin 0.10; weitere Bereinigung; LLL als nicht alleinige Source of Truth erkannt.

Result / significance: Grundsatz: möglichst ein positiver Spawn-Owner pro Enemy.

S1.11 — Erster Single-Owner-Versuch über LLL

Mehrere Enemies sollten zentral über LLL kontrolliert werden.

Result / significance: Für Rolling Giant/Siren Head später als unzuverlässig verworfen.

S1.12 — Vier-Enemy-Offense-Test

Nur Rolling Giant, Shy Guy, Locker innen und Siren Head draußen vorgesehen.

Result / significance: Observer leakte über zusätzliche Match-Regeln.

S1.13 — Runtime-Identifizierung/EnemyScan

Locker/Observer hart deaktiviert; Runtime-Namen getestet; EnemyScan installiert.

Result / significance: Don't Touch Me leakte; Rolling Giant Runtime-Weight 0; LLL-Ownership verworfen.

S1.14 — Native Spawn-Ownership

Rolling Giant20, Shy Guy7, SirenHead30 wieder native; konkurrierende LLL-Entries entfernt.

Result / significance: Praktisch bestätigt und weiterhin maßgeblich.

S1.15 — Rolling Giant juijui-Verhalten

RandomlyMoveWhileLooking, MoveSpeed2, Acceleration1 etc.

Result / significance: Bewährte Bewegungskonfiguration.

S1.16 — Random Enemy Sizes

RandomEnemiesSize 1.1.20 integriert.

Result / significance: Praktisch bestätigt.

S1.17 — Mirage/ShipWindows/Company-Vorbereitung

FearOverhauled entfernt; AutoCompanyBuilding 1.2.1; ShipWindows 2.11.1; LethalModDataLib entfernt.

Result / significance: AutoCompany später verworfen.

S1.18 — Company-Automation Test 1

AutoCompanyBuilding 1.2.1 beibehalten.

Result / significance: Auto-Routing nicht zuverlässig.

S1.19 — RandomMoonFX-Alternative

AutoCompanyBuilding entfernt; RandomMoonFX getestet.

Result / significance: Ungeeignet; nicht wiederverwenden.

S1.20 — AutoCompanyBuilding 1.1.3

Ältere Version getestet.

Result / significance: Routing teils möglich, Auto-Landung nicht; verworfen.

S1.21 — Pikmin Water Avoidance

Water Avoidance aktiviert; Company noch offen.

Result / significance: Löste Wasserproblem nicht.

S1.22 — CompanyBuildingEnhancements + Hold Scan

CompanyBuildingEnhancements 2.6.0 und LethalHUD Hold-to-Scan eingeführt.

Result / significance: In S1.23 bestätigt.

S1.23 — Pikmin Water Resistance + Company bestätigt

Alle Pikmin-Typen wasserresistent; Water Avoidance false; Deadline0→Gordion→Auto-Landung bestätigt.

Result / significance: Stabile Lösung.

S1.24 — Enemy Restore / breite Spawn-Basis

Breiter Enemy-Roster; Pikmin Speed1.3; Referenzverhältnisse etabliert.

Result / significance: Letzter extrem ausführlich dokumentierter Altstand.

S1.25 — SCP-999 Isolation

SCP-999 sollte nach der massiven SCP999AI/SnowyLib-NRE-Flut deaktiviert bleiben.

Result / significance: Zielzustand war SCP999 aus. Aktuelle S1.31–S1.34 Runtime-Logs zeigten später jedoch, dass dieser Zustand unbemerkt regressiert war; S1.36 korrigiert das erneut.

S1.26 — Spawn Balance

Immortal Snail max2; Screenshot-Verhältnisse bindend; Giant Sapsucker globaler Daytime-Ansatz.

Result / significance: Regeln für spätere Builds.

S1.27 — Equal Interior Rotation

Normale Interiors auf gleiche Baseline.

Result / significance: Vorbereitung 26er-Rotation.

S1.28 — Black Mesa Equal Rotation

Black Mesa über eigene DawnLib/Mod-Config Weight100 eingebunden.

Result / significance: Runtime: 26 viable, alle Weight100.

S1.29 — CodeRebirth Runtime Test

CodeRebirth1.6.9 + Abhängigkeiten integriert.

Result / significance: Normale Gameplay-Basis hinter S1.30.

S1.29D — Enemy Power Audit

S1.29 + RedPillEnemySpawn Diagnosemod.

Result / significance: Plugin lud, vollständige Runtime-Power-Tabelle blieb aus.

S1.30 — Power Caps / Mimicless / Pikmin Shield

Mimics+CoronerMimics entfernt; CR-Pikmin-Schalter false; höhere Indoor-Caps.

Result / significance: Flash Turret Schutz bestätigt; Indoor subjektiv etwas zu dicht.

S1.31 — Indoor Power Trim -4

Alle steuerbaren Indoor-Caps um4 reduziert; keine Weight-/Interior-/Pikmin-Änderungen.

Späterer S1.31-Test zeigte einen mehrminütigen Pikmin-Angriff auf LeafBoi(Clone). Derselbe Runtime-Stand zeigte außerdem überraschend wieder SCP999 2.4.0 plus Startup-NRE.

Result / significance: Letzte unveränderte Power-Cap-Basis, aber nicht mehr aktueller Gameplay-Stand.

S1.32 — Leaf Boy Blacklist + Mirage Keep Recordings

Auf Basis S1.31:

- Leaf boy an die bestehende LethalMin Attack Blacklist angehängt; die längere bestehende Liste blieb erhalten.
- Mirage neverDeleteRecordings=true gesetzt; andere beobachtete Mirage-Werte beibehalten.
- Mirage-Pfade geklärt: <Lethal Company>/Mirage/settings.json und <Lethal Company>/Mirage/Recording.

Runtime-Test: Beim Rückweg vom Dungeon war die Schiffshangartür geschlossen. Mit dem damaligen AJB-Mod sank die Türenergie nicht, sodass der einzige Spieler draußen dauerhaft ausgesperrt war.

Die konkrete ursprüngliche Close-Ursache wurde nicht bewiesen. Malfunctions zeigte keinen erfolgreichen relevanten Door-Event; BCMER war deaktiviert; ein Masked war zwar beim Schiff, aber Vanilla-Masked besitzt keine Hangar-Button-Interaktion.

Result / significance: Leaf/Mirage-Fix beibehalten. Ship-door-Lockout brauchte einen engeren Failsafe.

S1.33 — Erster Ship Door Failsafe

AJB-Keep_hangar_ship_door_closed deaktiviert. Eigene DLL sollte:

- bei lebendem Spieler im Schiff Türenergie auf 100% halten,
- bei allen lebenden Spielern draußen Vanilla-Drain erlauben,
- DoorAudit-Stacks loggen.

Result / significance: Später stellte sich heraus, dass die Custom-DLL beim normalen Gale-Import nicht übernommen/geladen worden war. Das Design war daher noch nicht runtime-getestet.

S1.34 — Malfunctions Disabled / Runtime-Test

zealsprince-Malfunctions auf ausdrücklichen Benutzerwunsch deaktiviert. Entscheidung: bleibt aus, bis der Benutzer irgendwann explizit Reaktivierung verlangt.

Runtime-Test:

- kein Custom-Door-Plugin Load-Eintrag,
- keine DoorAudit/DoorFailsafe-Zeilen,
- als einziger Spieler im gelandeten Schiff Tür geschlossen → Prozent sank und Tür öffnete bei 0%: reines Vanilla-Verhalten,
- enemies schien nicht alle tatsächlich gespawnten Enemies zu zeigen,
- Quellcodeprüfung von EnemyScan 1.2.1 bestätigte den ScanNode-Filter als Ursache,
- Puma identifiziert als Vanilla-Feiopar/PumaAI,
- Coin identifiziert als CodeRebirth-Währung für Denomination Analyzer / Merchant / Vending,
- SCP999 lud weiterhin und warf Startup-NRE.

Result / significance: Neuaufbau des lokalen Patches erforderlich; SCP999-Manifestregression erkannt.

S1.35 — Door + Complete Enemy Scan

Neue lokale S135CompatibilityFixes.dll gegen V81 GameLibs gebaut.

Enthält:

- robustere In-Ship-Erkennung über Vanilla-Flag plus shipInnerRoomBounds,
- Anti-Lockout-Failsafe über Vanilla-Hydraulik statt Force-Open-RPC,
- DoorAudit-Logging,
- EnemyScan-Patch ohne ScanNode-Voraussetzung für die Terminalanzeige,
- Malfunctions weiter deaktiviert,
- AJB weiter deaktiviert,
- Fallback local-mod ZIP.

Result / significance: Build/Archiv/Diff erfolgreich, aber noch kein Runtime-Test. Beim Handover-Audit wurde erkannt, dass SCP999 im Manifest noch aktiv war.

S1.36 — Handover Clean Baseline

S1.35 plus exakt eine Profilbereinigung:

- ProjectSCP-SCP999 von enabled auf disabled.

Kein anderer S1.35-Archivmember wurde geändert.

Manifest: 176 total / 170 active / 6 disabled.

Result / significance: Neuer kanonischer Übergabe-/Testkandidat. Erster nächster Test muss zuerst bestätigen, dass S135CompatibilityFixes.dll durch Gale tatsächlich importiert und von BepInEx geladen wird.